

Sakshi Giglani

☎ +91 7069052453 ✉ sakshigiglani27@gmail.com 🔗 linkedin.com/in/sakshi-giglani 🌐 github.com/027Sakshi

Education

Gujarat Law Society University, Gujarat

B.Tech. in Computer Engineering (Artificial Intelligence)

June 2023 – Present

CGPA: 7.85 / 10

Projects

LUGAI: Self-Healing Defense Against Adversarial Attacks

Aug 2025 – Dec 2025

- Built a real-time self-healing system to detect and recover deep learning models from adversarial attacks.
- Used dual detection signals based on prediction uncertainty and autoencoder reconstruction error.
- Implemented a denoising autoencoder pipeline that improved model accuracy from 22.97% to 93.02% after attack.
- Created an interactive Streamlit dashboard to visualize clean, attacked, and recovered samples with detection insights.

Image Caption Generation using CNN–LSTM with Attention

Oct 2025 – Nov 2025

- Developed an end-to-end image captioning system combining CNN-based visual feature extraction with LSTM-based language generation.
- Implemented an InceptionV3 encoder and attention-enabled LSTM decoder for context-aware caption generation.
- Trained the model on the MS COCO dataset with custom caption tokenization and sequence preprocessing.
- Evaluated caption quality on unseen images using BLEU score to measure semantic and linguistic accuracy.

AI-Powered Wellness & Skin Health Assistant

Sept 2025 – Present

- Built an AI-powered wellness assistant combining skin disease detection with personalized diet and exercise guidance.
- Implemented EfficientNet-B3 and MobileNetV2 models, achieving up to 80.40% validation accuracy.
- Applied two-stage transfer learning with dropout regularization to improve generalization and reduce overfitting.
- Deployed a real-time inference pipeline using FastAPI and Streamlit for scalable usage on standard hardware.

TradeGuard AI: Autonomous Risk Governance System

Jan 2026

- Built an autonomous AI system to govern real-time trade execution under behavioral and cognitive risk.
- Implemented modular agents for overtrading detection, stress scoring, market context, and mode inference.
- Designed a central risk governor to allow, delay, or block trades based on live trader state.
- Developed FastAPI-based decision and broker layers to enforce risk interventions safely.
- Created a Streamlit dashboard to visualize decisions, trader state, and system telemetry in real time.

Experience

Physical Research Laboratory (PRL), Ahmedabad

AI/ML Research Intern

May 2026 – Present

Ahmedabad, India

- Developed a vision-based framework for atmospheric visibility estimation from sky images.
- Implemented DINOv2 and XGBoost models for image-based visibility prediction.
- Applied atmospheric domain adaptation using public haze and local imagery datasets.
- Performed Aerosol Optical Depth (AOD) inference using physics-guided atmospheric relationships.

CSRBOX – IBM SkillsBuild (AICTE)

Applied AI Intern

Dec 2025 – Jan 2026

Remote

- Developed an AI-based skill-to-career mapping system to translate user skills into suitable job roles.
- Implemented NLP-based skill extraction, normalization, and confidence-based role ranking.
- Designed explainable AI outputs highlighting role suitability and skill gaps in human-readable form.
- Built a web-based application using Python, FastAPI, and Streamlit with downloadable analysis reports.

Technical Skills

Languages: Python, C, C++

Frameworks: PyTorch, TensorFlow, Streamlit, FastAPI

Libraries: NumPy, Pandas, Scikit-learn, OpenCV, NLTK, Matplotlib, Seaborn

Tools & Platforms: Git, GitHub, VS Code, Jupyter Notebook, Google Colab

Core Areas: Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, NLP, Generative AI, Data Visualization